

Statistical methods for computer science

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Discrete Probability Distributions: Bernoulli, Binomial, Poisson

Where were we?

Statistical science is the science of developing and applying methods for collecting, analyzing and interpreting data.

- ✓ Design phase
- ✓ Description phase

Inference:

- ✓ Probability
- ✓ Introduction to random variables
- ✓ Discrete and continuous random variables

Today Discrete Probability Distributions: Bernoulli, Binomial, Poisson

Course Outline

- Basic concepts
- Descriptive statistics (+ R Lab)
- Introduction to probability
- Introduction to random variables
- **Discrete probability distributions**
- Continuous probability distributions (+ R Lab)
- Sampling distributions
- Statistical inference: estimation (+ R Lab)
- Statistical inference: testing (+ R Lab)
- Linear model: estimation, inference, diagnostics (+ R Lab)

Bernoulli distribution

- 1 Practical applications
- 2 PMF
- 3 Main expectations (mean, variance)

Practical Applications

It is used to describe binary phenomena, i.e., having only two possible outcomes coded by 1 and 0.

Flipping a coin: “head”
1, “tail” 0



Outcome of an exam:
“pass” 1, “fail” 0



Weather forecast:
“sun” 1, “rain” 0



PMF

X is a Bernoulli RV iff its PMF is given by

$$f_X(x) = P(X = x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

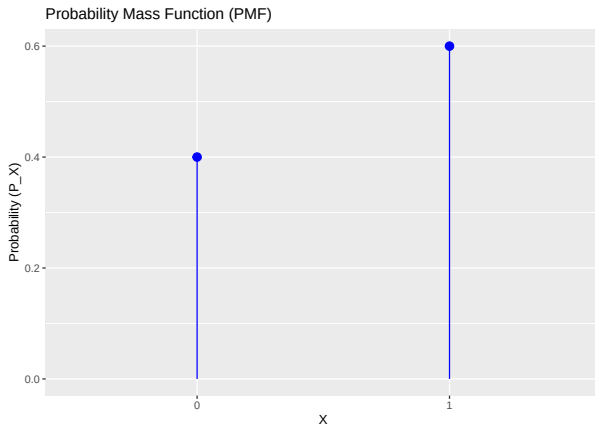
Remarks

- $X \sim Be(p)$ indicates that X is distributed as a Bernoulli with parameter p , the probability of “success”
- Domain: $\mathcal{D}_f = \{0, 1\}$, parameter space: $p \in [0, 1]$
- The probability of failure is often denoted as $q = 1 - p$
- The PMF can be written in a more compact way as follows:

$$f_X(x) = \begin{cases} p^x q^{1-x} & \text{if } x \in \{0, 1\} \\ 0 & \text{otherwise} \end{cases}$$

PMF

The PMF of a Bernoulli RV can be visualized by the usual plot



Main expectations

Let $X \sim Be(p)$, its expected value is given by $E[X] = p$ and the variance is given by $V[X] = pq$.

Proof.

By the definition of the expected value of a discrete RV,

$$E[X] = \sum_x x \cdot f_X(x) = 1 \cdot p + 0 \cdot q = p$$

The second moment of a discrete RV is

$$E[X^2] = \sum_x x^2 \cdot f_X(x) = 1^2 \cdot p + 0^2 \cdot q = p$$

By the shortcut for the variance,

$$V[X] = E[X^2] - E[X]^2 = p - p^2 = p(1 - p) = pq$$

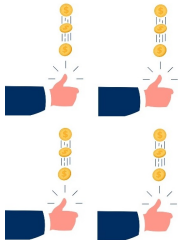
Binomial distribution

- 1 Practical applications
- 2 PMF
- 3 Main expectations (mean, variance)
- 4 Main properties

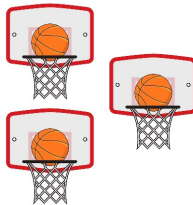
Practical Applications

It is used to describe the number of successes in a number of independent repetitions of a Bernoulli experiment

Repeated coin flipping



Basketball throw attempts



PMF

X is a Binomial RV iff its PMF is given by

$$f_X(x) = P(X = x) = \begin{cases} \binom{n}{x} p^x q^{n-x} & \text{if } x \in \mathbb{N} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Remarks

- $X \sim Bi(n, p)$ indicates that X is distributed as a Binomial with parameters n, p
- Domain $\mathcal{D}_f = \mathbb{N}$, parameter space $n \in \mathbb{N}, p \in [0, 1]$
- $\binom{n}{x} = C_{n,x} = \frac{n!}{x!(n-x)!}$
- The factorial operator is defined as
$$\begin{cases} x! = x(x-1)(x-2) \cdots 1 \\ 0! = 1 \end{cases}$$

PMF

To understand the above formula for the PMF let's consider an example: compute the probability of getting 5 heads in 8 independent coin flips. By formula (2) we have:

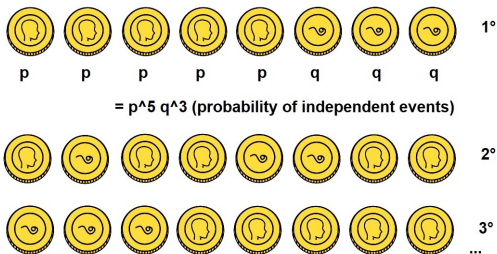
$$P(X = 5) = \binom{8}{5} p^5 q^3 = 56 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^3$$

Why?

Define X_i a RV that is 1 when we get a “head” in the i -th coin toss (e.g., X_3 takes value 1 if we get head at the 3rd coin flip). Imagine to flip the coin 8 times and getting this sequence: (1, 1, 1, 1, 1, 0, 0, 0). Its probability is given by

$$\begin{aligned} P(Y = 5) &= P(X_1 = 1, \dots, X_5 = 1, X_6 = 0, X_7 = 0, X_8 = 0) \\ &= P(X_1 = 1) \cdot \dots \cdot P(X_5 = 1) P(X_6 = 0) \cdot \dots \cdot P(X_8 = 0) \\ &= p \cdot p \cdot p \cdot p \cdot p \cdot q \cdot q \cdot q = p^5 q^3 \end{aligned}$$

This is just the probability of a given sequence, but there are other ways to obtain 5 successes in 8 trials:

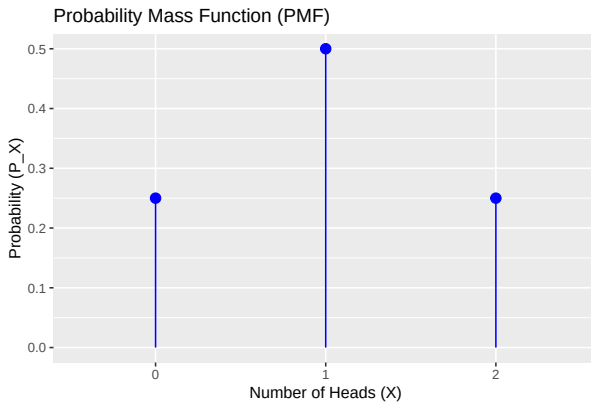


These are the permutations of 8 elements with repetitions: p repeats $n_1 = 5$ times and q repeats $n_2 = 3$ times. We know from combinatorial mathematics that $P_n^{n_1, n_2} = \frac{n!}{n_1! n_2!} = \frac{8!}{5! 3!} = 56$. This is also equal to the number of ways we can obtain k successes in n trials, i.e.,

$$C_{n,k} = \frac{n!}{k!(n-k)!} = \frac{8!}{5!3!}$$

PMF

The PMF of a Binomial RV can be visualized by the usual plot



Main properties

Link with Bernoulli I

① $Bi(1, p) = Be(p)$

Check this by imposing $n = 1$ in the Binomial PMF

Link with Bernoulli II

② $(X_1, X_2, \dots, X_n) \sim i.i.d Be(p) \Rightarrow \sum_{i=1}^n X_i \sim Bi(n, p)$

Additivity

③ $X_1 \sim Bi(n_1, p), X_2 \sim Bi(n_2, p)$ and
 $X_1 \perp X_2 \Rightarrow X_1 + X_2 \sim Bi(n_1 + n_2, p)$

Remark: the additivity property only works when the RVs have the same parameter p

Main expectations

Let $X \sim \text{Bi}(n, p)$, its expected value is given by $E[X] = np$ and the variance is given by $V[X] = npq$.

Proof.

Compute the expectation of $Y = \sum_{i=1}^n X_i$ where $X_i \sim i.i.d \text{Be}(p)$,

$$\begin{aligned} E[Y] &= E[X_1 + \dots + X_n] = E\left[\sum_{i=1}^n X_i\right] \\ &= \sum_{i=1}^n E[X_i] = \sum_{i=1}^n p = np \end{aligned}$$

By the same token and since the X_i 's are independent,

$$\begin{aligned} V[X] &= V\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n V[X_i] \\ &= \sum_{i=1}^n pq = npq \end{aligned}$$

Exercise

You work in a quality control department at computer chip manufacturing company. Each chip has a 5% chance of being defective and you randomly select 10 chips for inspection.

- ① What is the probability that exactly 2 chips out of the 10 are defective?
- ② What is the probability that at least 1 chip in the sample is defective?

Exercise

You work in a quality control department at computer chip manufacturing company. Each chip has a 5% chance of being defective and you randomly select 10 chips for inspection.

- 1 What is the probability that exactly 2 chips out of the 10 are defective?
- 2 What is the probability that at least 1 chip in the sample is defective?

Let X be a RV counting the number of defective computer chips among the 10 that have been sampled.

$$1 \quad P(X = 2) = \binom{10}{2} 0.05^2 \cdot 0.95^8 = 7.46\%$$

$$2 \quad P(X \geq 1) = 1 - P(X < 1) = 1 - P(X = 0) = 1 - \binom{10}{0} 0.05^0 \cdot 0.95^{10} = \\ = 1 - 0.5987 = 40.12\%$$

Poisson distribution

- 1 Practical applications
- 2 PMF
- 3 Main expectations (mean, variance)
- 4 Main properties

Practical Applications

It is used to describe the number of times an event happens in a fixed portion of time or in a fixed portion of space

Number of cars arriving
in a parking lot in 1 hour



Number of calls to a call
center in 1 min



Number of pine trees
in a forest



PMF

X is a Poisson RV iff its PMF is given by

$$f_X(x) = P(X = x) = \begin{cases} \frac{\lambda^x e^{-\lambda}}{x!} & \text{if } x \in \mathbb{N} \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

Remarks:

- $X \sim Po(\lambda)$ indicates that X is distributed as a Poisson with parameter $\lambda > 0$, fixed rate of occurrence
- Domain $\mathcal{D} = \mathbb{N}$, parameter space $\lambda \in (0, +\infty)$
- A Poisson RV must satisfy three additional assumptions: i) events occur independently ; ii) constant rate of occurrence; iii) the events cannot occur simultaneously

Main expectations

Let $X \sim Po(\lambda)$, its expected value and variance are given by $E[X] = V[X] = \lambda$.

Proof.

$$\begin{aligned} E[X] &= \sum_{x=0}^{\infty} x \frac{\lambda^x e^{-\lambda}}{x!} = \sum_{x=0}^{\infty} x \frac{\lambda \lambda^{x-1} e^{-\lambda}}{x(x-1)!} = e^{-\lambda} \lambda \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} \\ &= e^{-\lambda} \lambda \sum_{z=0}^{\infty} \frac{\lambda^z}{z!} = e^{-\lambda} \lambda e^{\lambda} = \lambda \end{aligned}$$

$$\begin{aligned} E[X^2] &= \sum_{x=1}^{\infty} x^2 \frac{\lambda^x e^{-\lambda}}{x!} = e^{-\lambda} \lambda \sum_{x=1}^{\infty} (x-1+1) \frac{\lambda^{x-1}}{(x-1)!} \\ &= e^{-\lambda} \lambda^2 \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} + e^{-\lambda} \lambda \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} = \lambda^2 + \lambda \end{aligned}$$

$$V[X] = E[X^2] - E[X]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

Main Properties

Binomial Approximation

- ① $Bi(n, p) \approx Po(\lambda = np)$ when $p \rightarrow 0$ and $n \rightarrow \infty$

Remark: for the approximation to work, the expectations of the two RVs must match (at least approximately), so

- $E[X] = np = \lambda$
- $V[X] = npq \simeq \lambda$, for this to hold (approximately) $q \rightarrow 1$ which means $p \rightarrow 0$
- good approximation when $n \geq 20, p \leq 0.05$, excellent approximation when $n \geq 100, p \leq 0.1$

Main Properties

Additivity

2 $X \sim Po(\lambda_1)$, $Y \sim Po(\lambda_2)$ and $X \perp Y \Rightarrow Z = X + Y \sim Po(\lambda_1 + \lambda_2)$

Proof.

Joint PMF:

$$P(Z = n, X = x) = \frac{\lambda_1^x e^{-\lambda_1}}{x!} \frac{\lambda_2^{n-x} e^{-\lambda_2}}{(n-x)!} = \frac{\lambda_1^x \lambda_2^{n-x} e^{-(\lambda_1 + \lambda_2)}}{x!(n-x)!}$$

Marginal PMF:

$$\begin{aligned} P(Z = n) &= \sum_{x=0}^n \frac{\lambda_1^x e^{-\lambda_1}}{x!} \frac{\lambda_2^{n-x} e^{-\lambda_2}}{(n-x)!} = \frac{e^{-(\lambda_1 + \lambda_2)}}{n!} \sum_{x=0}^n \binom{n}{x} \lambda_1^x \lambda_2^{n-x} \\ &= \frac{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^n}{n!} \end{aligned}$$

Main Properties

Link with Binomial

- 3 $X \sim Po(\lambda_1)$, $Y \sim Po(\lambda_2)$ and $X \perp Y \Rightarrow X|Z \sim Bi(n, p)$ with $p = \lambda_1/(\lambda_1 + \lambda_2)$

Proof.

$$\begin{aligned} f_{X|Z}(x|z) &= \frac{P(Z = n, X = x)}{P(Z = n)} = \frac{\lambda_1^x \lambda_2^{n-x} n!}{x!(n-x)!(\lambda_1 + \lambda_2)^n} \\ &= \binom{n}{x} \frac{\lambda_1^x}{(\lambda_1 + \lambda_2)^x} \frac{\lambda_2^{n-x}}{(\lambda_1 + \lambda_2)^{n-x}} = \binom{n}{x} p^x (1-p)^{n-x} \end{aligned}$$

which means that $X|Z \sim Bi(n, p)$ with $p = \lambda_1/(\lambda_1 + \lambda_2)$.

Exercise

Between January 2001 and February 2017, there have been 38 earthquakes within 50km from Florence measuring 3.0 or more on the Richter scale (source: [INGV](#)).

- 1 What is the probability of having exactly three earthquakes of magnitude greater or equal 3.0 in this year?
- 2 What is the probability of having at least two earthquakes?
- 3 Is the Binomial distribution a good approximation to the Poisson distribution in this example? Motivate and, if so, calculate the probability of having exactly three earthquakes in a year using the Binomial distribution
- 4 Plot the PMF and the CDF (for $0 \leq x \leq 5$)

Exercise

- ① Let X be a RV describing the number of earthquakes in a year. The rate of occurrence is $\lambda = 38/16 = 2.38$. Under the assumptions that the earthquakes are independent and cannot occur simultaneously, $X \sim Po(\lambda)$,

$$P(X = 3) = \frac{\lambda^x e^{-\lambda}}{x!} = \frac{2.38^3 e^{-2.38}}{3!} = 20.80\%$$

- ② Using the same formula,

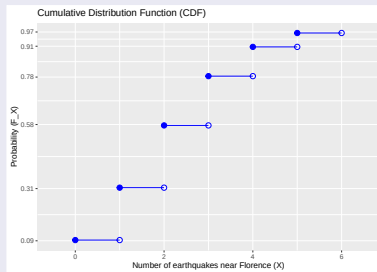
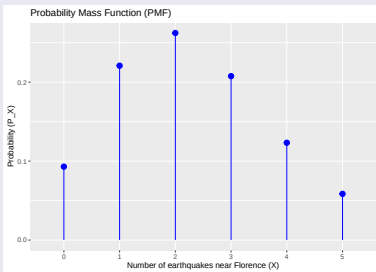
$$\begin{aligned} P(X \geq 2) &= 1 - P(X < 2) = 1 - (P(X = 0) + P(X = 1)) \\ &= 1 - \frac{2.38^0 e^{-2.38}}{0!} - \frac{2.38^1 e^{-2.38}}{1!} = 68.72\% \end{aligned}$$

- ③ Let Y be a RV counting the number of successes (earthquakes) in $n = 365$ trials (1 year). We know there are $\lambda = 2.38 = np$ earthquakes in a year. Thus, the probability of “success” is $p = 2.38/365 = 0.0065$. Since $n = 365 > 100$ and $p = 0.0065 < 0.1$ the Binomial is a good approximation to the Poisson.

$$P(Y = 3) = \frac{365!}{362!3!} (0.0065)^3 (0.9935)^{362} = 20.82\%$$

Exercise

4 Plotting the PMF and the CDF



Exercise

An hotel has two distinct online-based booking platforms, A and B. Platform A receives requests at a rate of 1 every 4 minutes while platform B receives requests at a rate of 1 every 2 minutes. Find:

- 1 Probability that in 1 minute platform A receives 2 or more requests
- 2 Probability of not receiving requests for 10 minutes for platform B
- 3 Probability that the hotel receives 4 bookings in a minute coming from both platforms
- 4 Probability that out of these 4 bookings, 3 come from platform B

Let X (Y) be the RV counting the number of requests received by platform A (B) in a minute. Under the assumptions that the requests are independent, they cannot be simultaneous and that the rate is constant, X (Y) can be modeled with a Poisson distribution.

- ① $\lambda_A = 1/4$ requests/min.

$$P(X > 1) = 1 - P(X = 0) - P(X = 1) = 1 - e^{-0.25} - \lambda e^{-0.25} = 2.65\%$$

- ② $\lambda_B = 1/2$ requests/min, meaning that the rate of requests in 10 minutes is $\lambda_B^{(10)} = 5$. Thus,

$$P(Y = 0) = e^{-5} = 0.67\%$$

- ③ $\lambda = \lambda_A + \lambda_B = 3/4$ requests/min in total from both platforms. Thus,

$$P(X + Y = 4) = \frac{0.75^4 e^{-0.75}}{4!} = 0.62\%$$

- ④ $Y|Z = n \sim Bi(n, p)$ with $Z = X + Y$ and $p = \lambda_B/\lambda$, then

$$P(Y = 3|Z = 4) = \binom{4}{3} p^3 (1 - p) = 39.51\%$$